- Cell wall resists internal turgor pressure of cell.
- Cell wall regulates growth rate and diffusion of materials.
- Cell wall functions as stores of carbohydrates.
- Cell wall protects against pathogens, dehydration, and other environmental factors.

1.4.3.Nucleus

The largest organelle in the cell is nucleus which is enveloped by bound double layered nuclear membrane preserving the genetic material called chromatin. Nucleus occupies 1%-2% and 10% in yeast and animal cells respectively. The genetic material forms a mass called chromatin that is concentrated in one part of the nucleus. The outer and inner membranes are separated by lumen. The outer membrane of the nuclear envelope is continuous with the endoplasmice reticulum (ER) membrane, and the lumen of the nuclear envelope is continuous with the lumen of the ER. The inner nuclear membrane is usually supported by a network of filaments called the nuclear lamina, located in the nucleus and anchored to the inner membrane. The nucleus contains subcompartments with specialized functions and the major subcompartment in the nucleus is the nucleolus.

The pores of nuclear membranes are large enough to be completely permeable to smaller molecules, so there is no difference in the aqueous environment of the nucleus and the cytoplasm. The nucleus is considered to be the core of the cell which regulates all metabolic events.

DO YOU KNOW?

The diversity of eukaryotic organisms with their nuclei extends from single cells whose existences resemble those of free-living bacteria to complex multicellular organisms with many different types of component cells. In the nucleus, the concentration of DNA is equivalent to a gel of high viscosity. In the other compartments, proteins are concentrated at a high density.

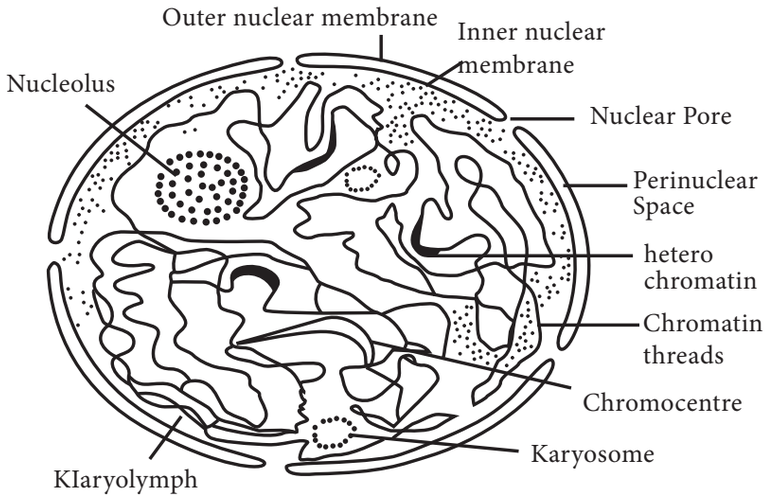


Figure 1.6 A diagrammatic representation of the ultrastructure of nucleus

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